Causal Analysis Report

BPI_Challenge_2013_closed_problems.xes.gz-flw

Decision points: 8147

Choice sets: 1

Transitions fitted: 7

Significant (p<0.05): 0

Generated: 2026-06-19 14:12

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

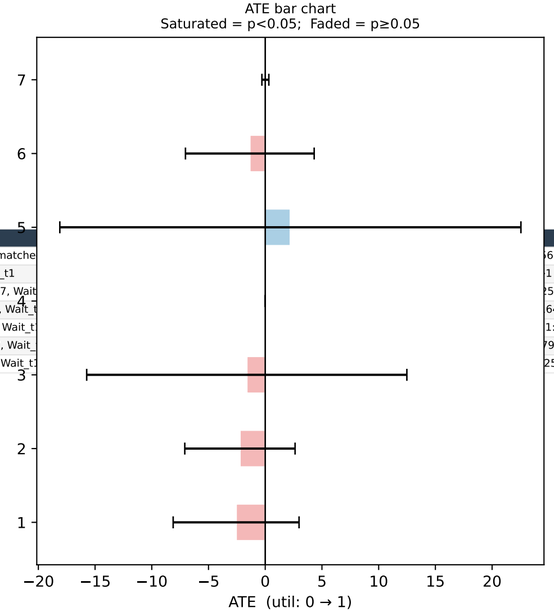
(no data)

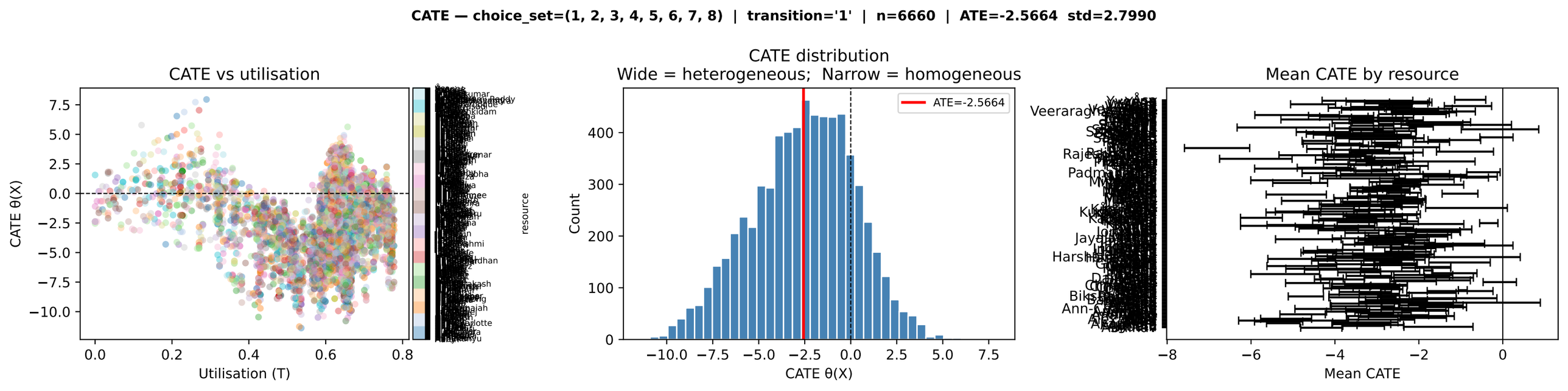
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

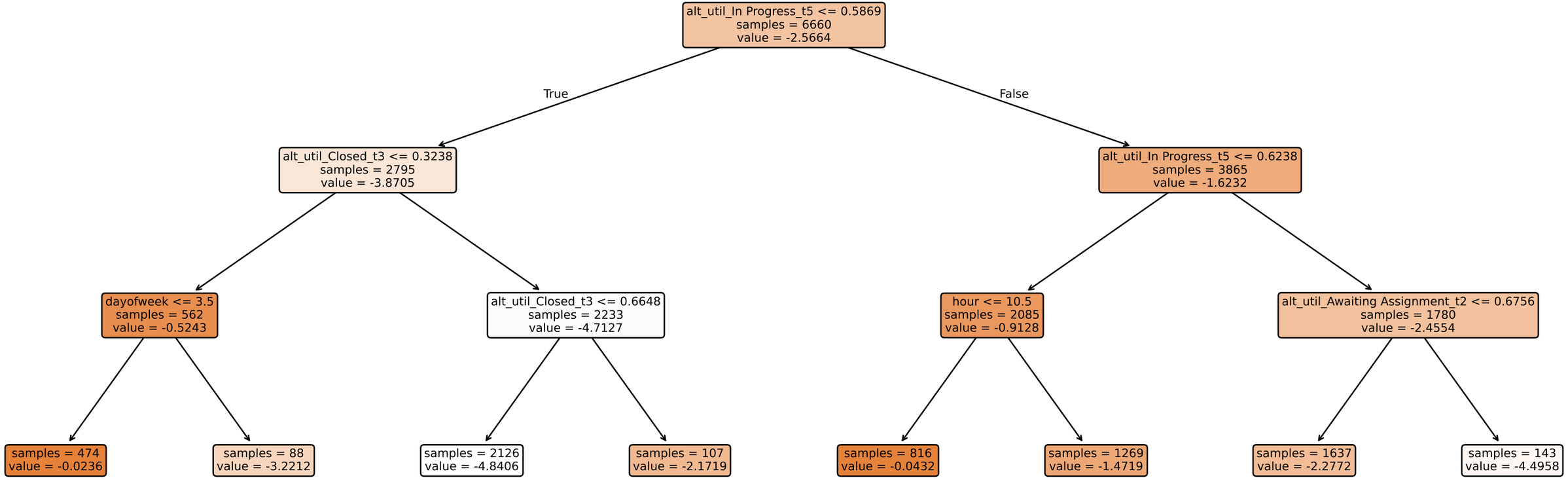
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	1	Wait_t1	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Unmatched_t7	-0.776369	0.207513	[-1.1831, -0.3696]	0.0002	True	θ=-0.7764: 1-unit ↑ util ↓ P(t1/Wait_t1) by 77.6%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	2	Awaiting Assignment_t2	Assigned_t6, Cancelled_t4, Closed_t3, In Progress_t5, Unmatched_t7, Wait_t1	0.247187	0.132698	[-0.0129, 0.5073]	0.0625	False	θ=+0.2472: 1-unit ↑ util ↑ P(t2/Awaiting Assignment_t2) by 24.7%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	3	Closed_t3	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, In Progress_t5, Unmatched_t7, Wait_t1	5.08511	0.499665	[4.1058, 6.0645]	0.0	True	θ=+5.0851: 1-unit ↑ util ↑ P(t3/Closed_t3) by 508.5%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	4	Cancelled_t4	Assigned_t6, Awaiting Assignment_t2, Closed_t3, In Progress_t5, Unmatched_t7, Wait_t1	-0.001004	0.00174	[-0.0044, 0.0024]	0.5638	False	θ=-0.0010: 1-unit ↑ util ↓ P(t4/Cancelled_t4) by 0.1%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	5	In Progress_t5	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, Unmatched_t7, Wait_t1	6.479506	0.596124	[5.3111, 7.6479]	0.0	True	θ=+6.4795: 1-unit ↑ util ↑ P(t5/In Progress_t5) by 648.0%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	6	Assigned_t6	Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Unmatched_t7, Wait_t1	-0.150351	0.231755	[-0.6046, 0.3039]	0.5165	False	θ=-0.1504: 1-unit ↑ util ↓ P(t6/Assigned_t6) by 15.0%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	7	Unmatched_t7	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Wait_t1	-0.006332	0.007416	[-0.0209, 0.0082]	0.3932	False	θ=-0.0063: 1-unit ↑ util ↓ P(t7/Unmatched_t7) by 0.6%

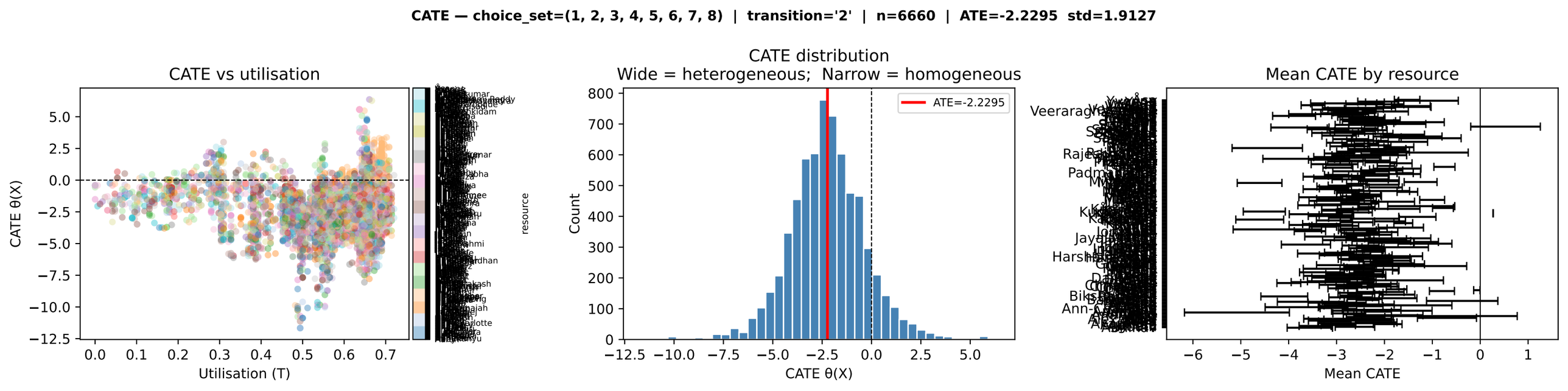
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

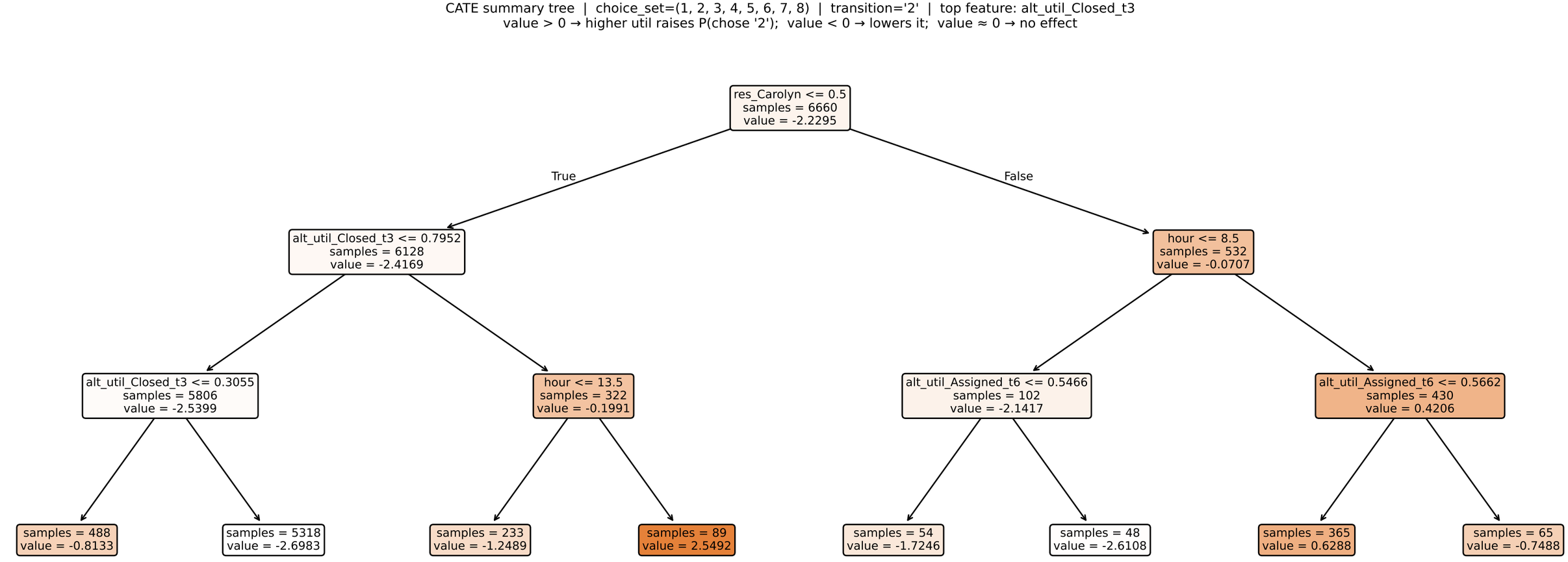
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	note
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	1	Wait_t1	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Unmatche	664: 0→1 util ↓ P(t1/Wait_t1) by 256.6%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	2	Awaiting Assignment_t2	Assigned_t6, Cancelled_t4, Closed_t3, In Progress_t5, Unmatched_t7, Wait_t1	1 util ↓ P(t2/Awaiting Assignment_t2) by 222.9%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	3	Closed_t3	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, In Progress_t5, Unmatched_t7, Wait	257: 0→1 util ↓ P(t3/Closed_t3) by 162.6%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	4	Cancelled_t4	Assigned_t6, Awaiting Assignment_t2, Closed_t3, In Progress_t5, Unmatched_t7, Wait_t	64: 0→1 util ↓ P(t4/Cancelled_t4) by 1.6%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	5	In Progress_t5	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, Unmatched_t7, Wait_t	1: 0→1 util ↑ P(t5/In Progress_t5) by 221.7%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	6	Assigned_t6	Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Unmatched_t7, Wait_	79: 0→1 util ↓ P(t6/Assigned_t6) by 135.8%
(1, 2, 3, 4, 5, 6, 7, 8)	('Wait_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'Cancelled_t4', 'In Progress_t5', 'Assigned_t6', 'Unmatched_t7', 'τ_t8')	6660	1, 2, 3, 4, 5, 6, 7	7	Unmatched_t7	Assigned_t6, Awaiting Assignment_t2, Cancelled_t4, Closed_t3, In Progress_t5, Wait_t	25: 0→1 util ↑ P(t7/Unmatched_t7) by 1.3%

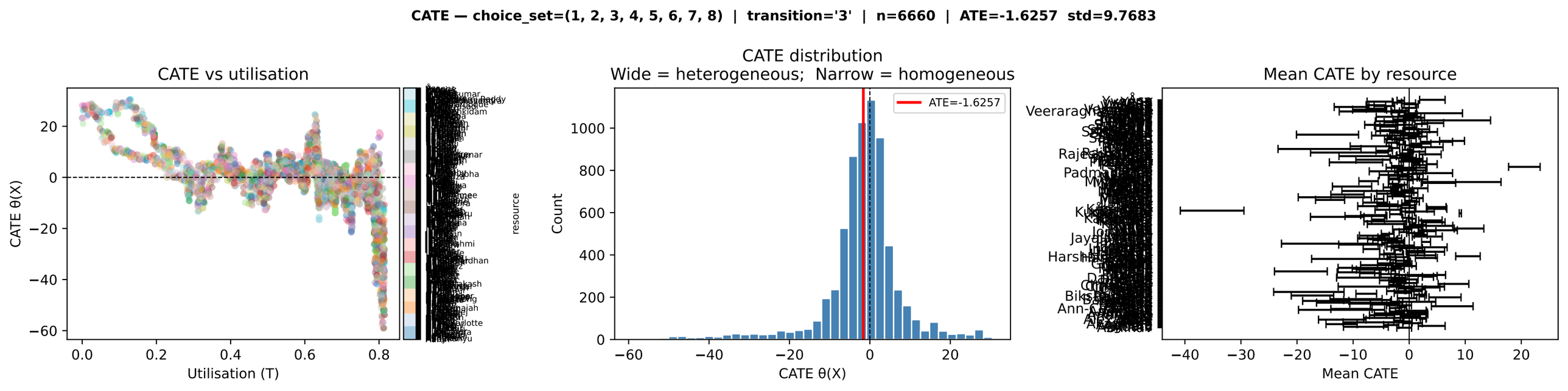


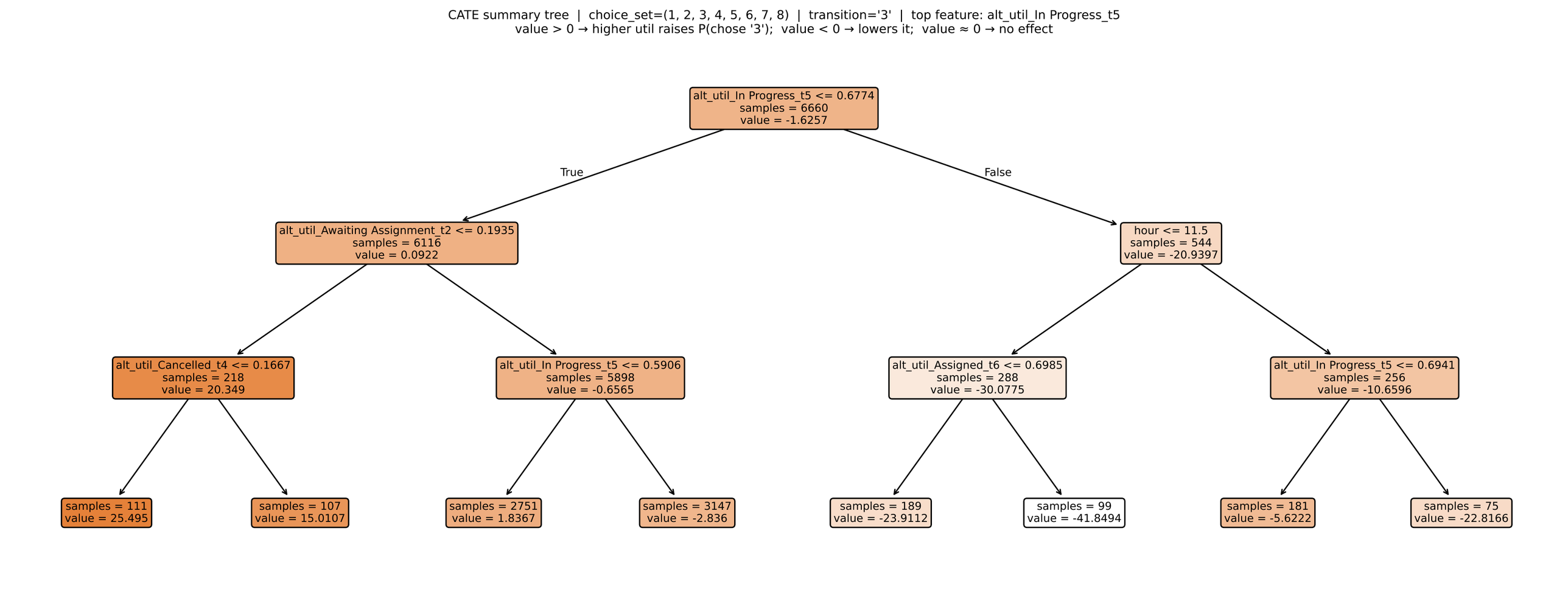

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='1' | n=6660 | ATE=-2.5664 std=2.7990
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.5664
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

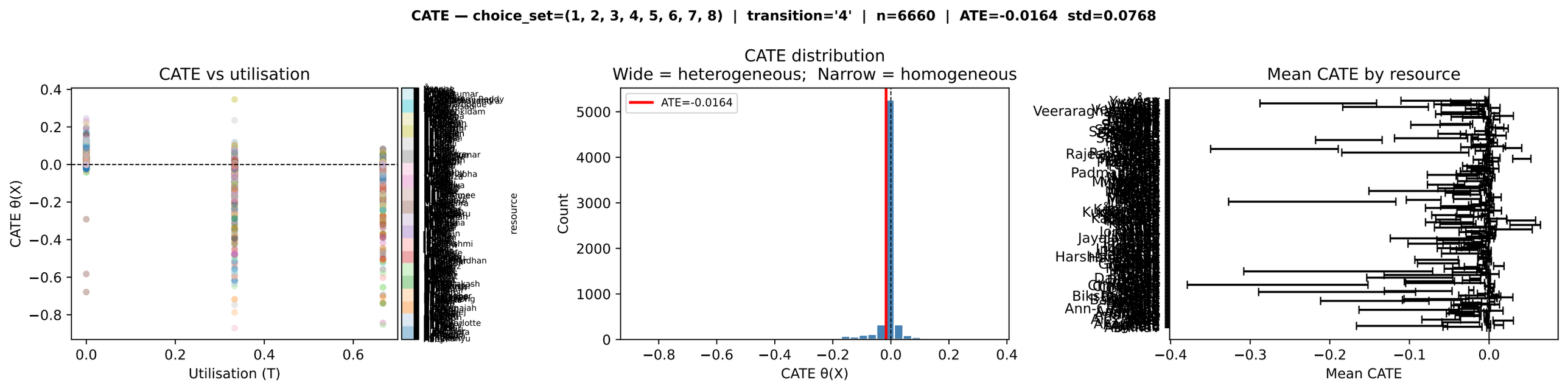

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='1' | top feature: alt_util_ln Progress_t5
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t5 <= 0.5869
samples = 6660
value = -2.5664
True
False
alt_util_Closed_t3 <= 0.3238
samples = 2795
value = -3.8705
alt_util_ln Progress_t5 <= 0.6238
samples = 3865
value = -1.6232
dayofweek <= 3.5
samples = 562
value = -0.5243
alt_util_Closed_t3 <= 0.6648
samples = 2233
value = -4.7127
hour <= 10.5
samples = 2085
value = -0.9128
alt_util_Awaiting Assignment_t2 <= 0.6756
samples = 1780
value = -2.4554
samples = 474
value = -0.0236
samples = 88
value = -3.2212
samples = 2126
value = -4.8406
samples = 107
value = -2.1719
samples = 816
value = -0.0432
samples = 1269
value = -1.4719
samples = 1637
value = -2.2772
samples = 143
value = -4.4958

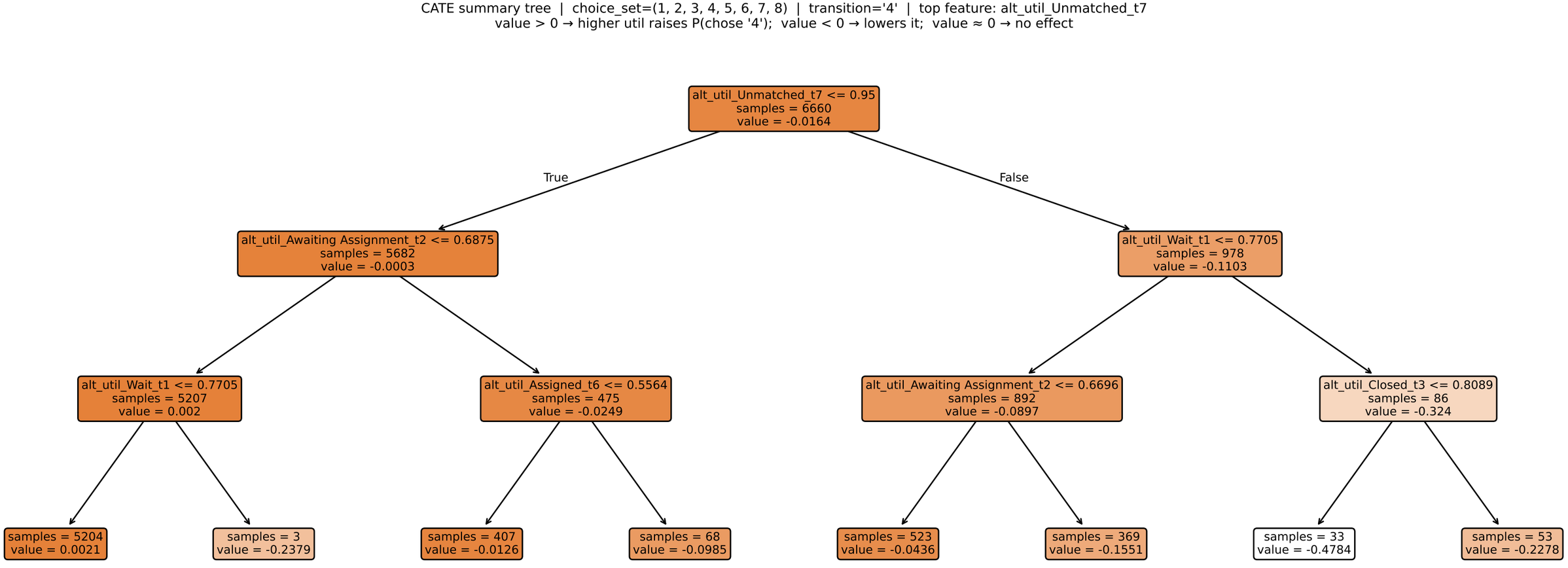

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='2' | n=6660 | ATE=-2.2295 std=1.9127
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.2295
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='2' | top feature: alt_util_Closed_t3
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
res_Carolyn <= 0.5
samples = 6660
value = -2.2295
True
False
alt_util_Closed_t3 <= 0.7952
samples = 6128
value = -2.4169
hour <= 8.5
samples = 532
value = -0.0707
alt_util_Closed_t3 <= 0.3055
samples = 5806
value = -2.5399
hour <= 13.5
samples = 322
value = -0.1991
alt_util_Assigned_t6 <= 0.5466
samples = 102
value = -2.1417
alt_util_Assigned_t6 <= 0.5662
samples = 430
value = 0.4206
samples = 488
value = -0.8133
samples = 5318
value = -2.6983
samples = 233
value = -1.2489
samples = 89
value = 2.5492
samples = 54
value = -1.7246
samples = 48
value = -2.6108
samples = 365
value = 0.6288
samples = 65
value = -0.7488

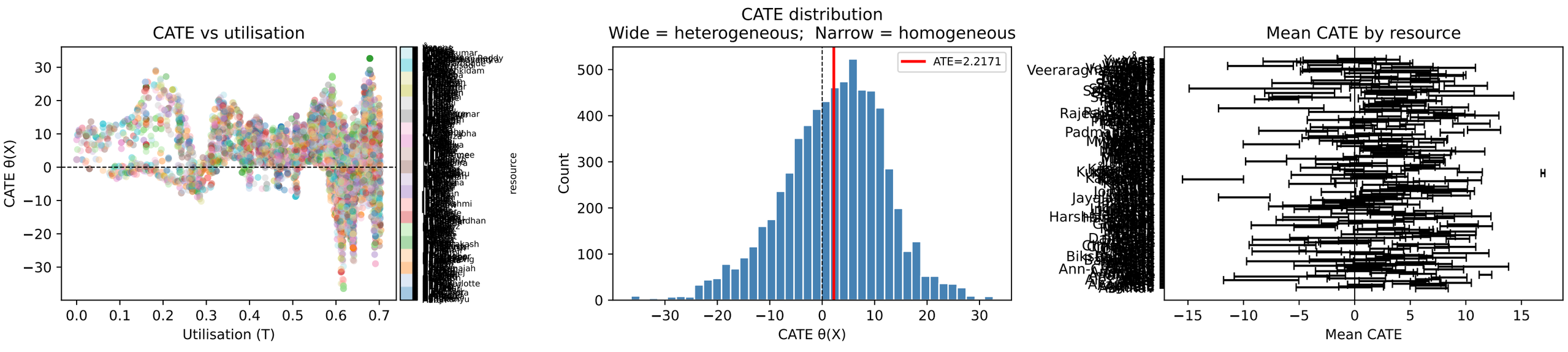

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='3' | n=6660 | ATE=-1.6257 std=9.7683
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.6257
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

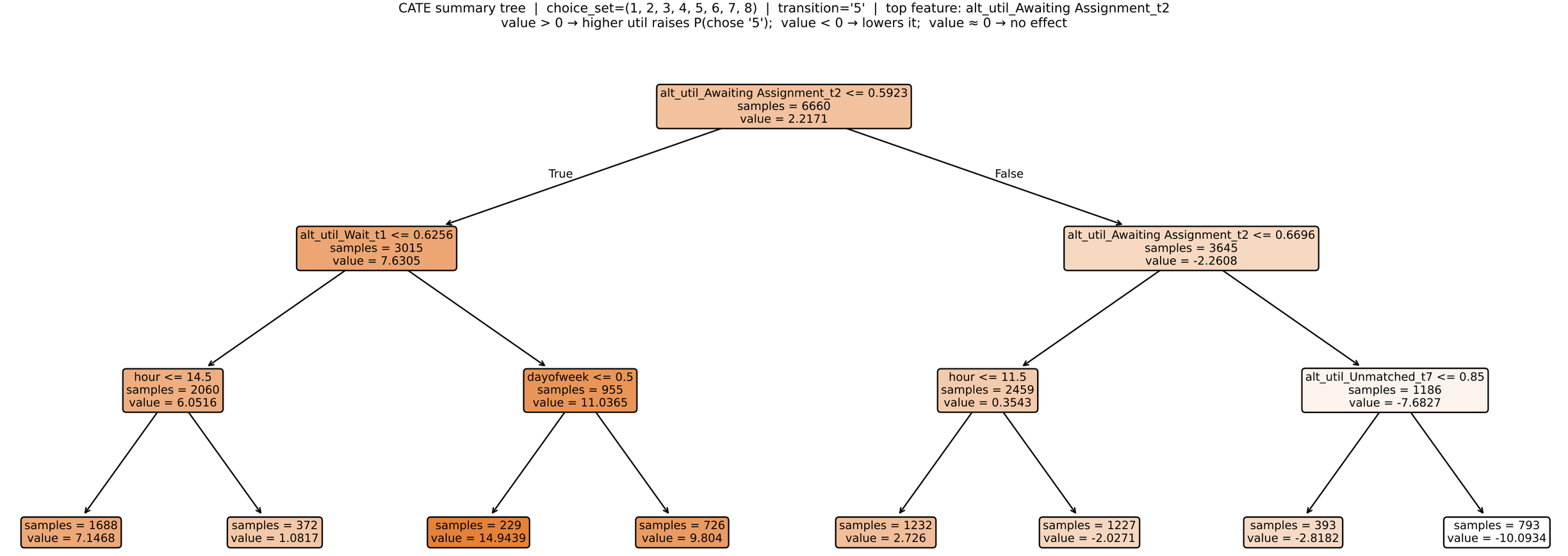

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='3' | top feature: alt_util_ln Progress_t5
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t5 <= 0.6774
samples = 6660
value = -1.6257
True
False
alt_util_Awaiting Assignment_t2 <= 0.1935
samples = 6116
value = 0.0922
hour <= 11.5
samples = 544
value = -20.9397
alt_util_Cancelled_t4 <= 0.1667
samples = 218
value = 20.349
alt_util_ln Progress_t5 <= 0.5906
samples = 5898
value = -0.6565
alt_util_Assigned_t6 <= 0.6985
samples = 288
value = -30.0775
alt_util_ln Progress_t5 <= 0.6941
samples = 256
value = -10.6596
samples = 111
value = 25.495
samples = 107
value = 15.0107
samples = 2751
value = 1.8367
samples = 3147
value = -2.836
samples = 189
value = -23.9112
samples = 99
value = -41.8494
samples = 181
value = -5.6222
samples = 75
value = -22.8166

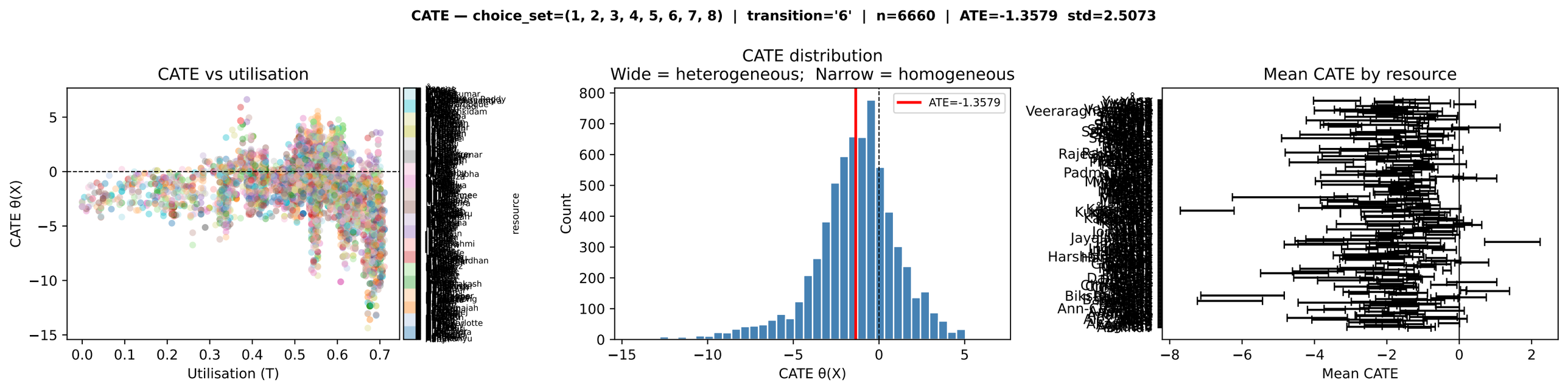

CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='4' | n=6660 | ATE=-0.0164 std=0.0768
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0164
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

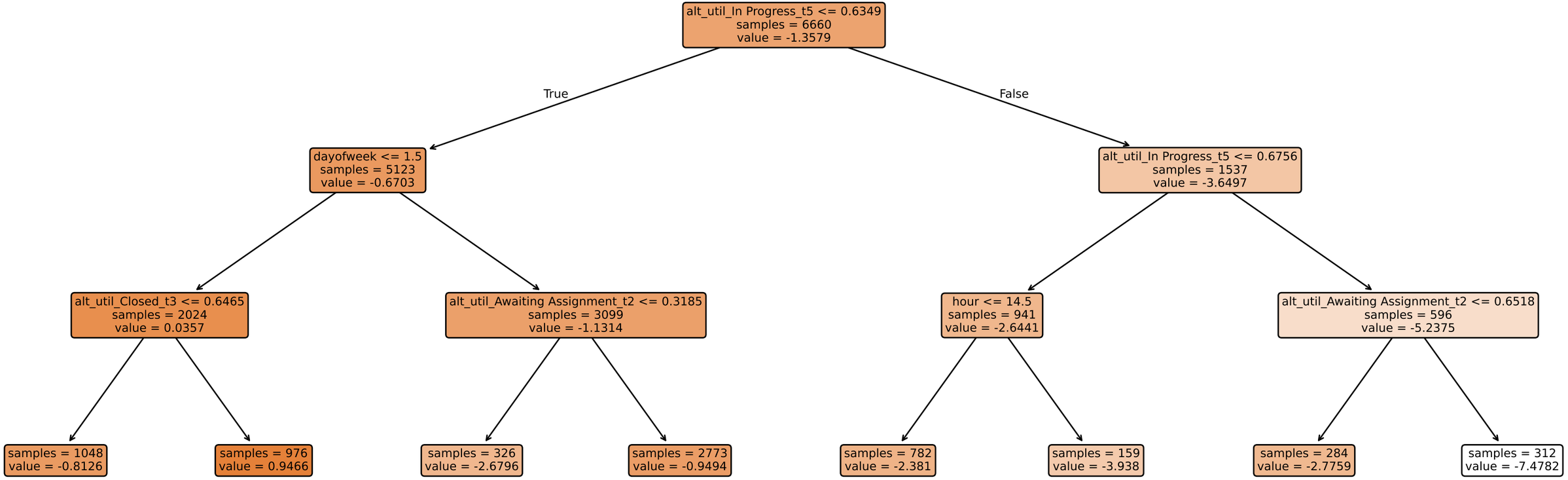

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='4' | top feature: alt_util_Unmatched_t7
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Unmatched_t7 <= 0.95
samples = 6660
value = -0.0164
True
False
alt_util_Awaiting Assignment_t2 <= 0.6875
samples = 5682
value = -0.0003
alt_util_Wait_t1 <= 0.7705
samples = 978
value = -0.1103
alt_util_Wait_t1 <= 0.7705
samples = 5207
value = 0.002
alt_util_Assigned_t6 <= 0.5564
samples = 475
value = -0.0249
alt_util_Awaiting Assignment_t2 <= 0.6696
samples = 892
value = -0.0897
alt_util_Closed_t3 <= 0.8089
samples = 86
value = -0.324
samples = 5204
value = 0.0021
samples = 3
value = -0.2379
samples = 407
value = -0.0126
samples = 68
value = -0.0985
samples = 523
value = -0.0436
samples = 369
value = -0.1551
samples = 33
value = -0.4784
samples = 53
value = -0.2278

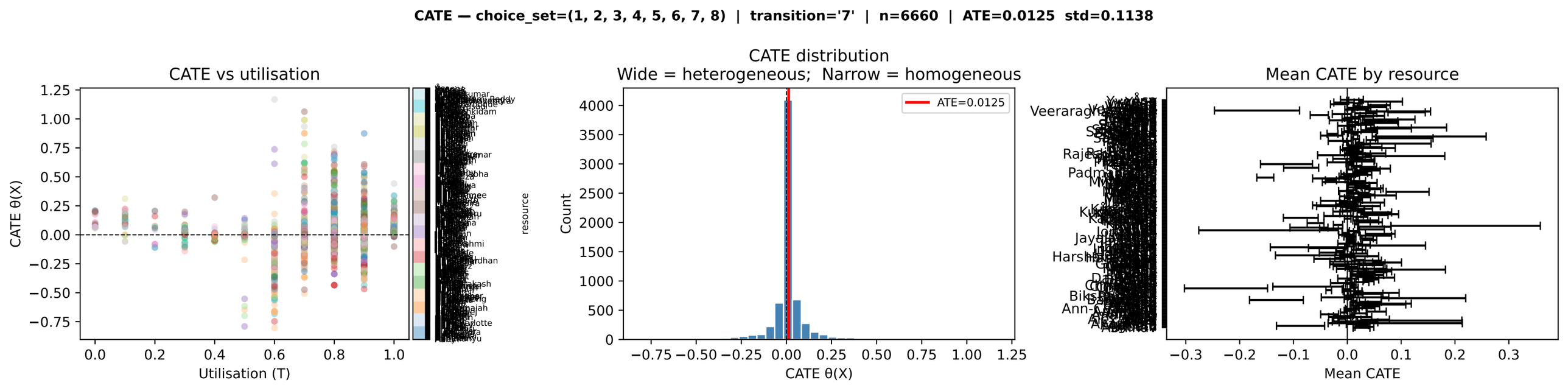
CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='5' | n=6660 | ATE=2.2171 std=9.8057

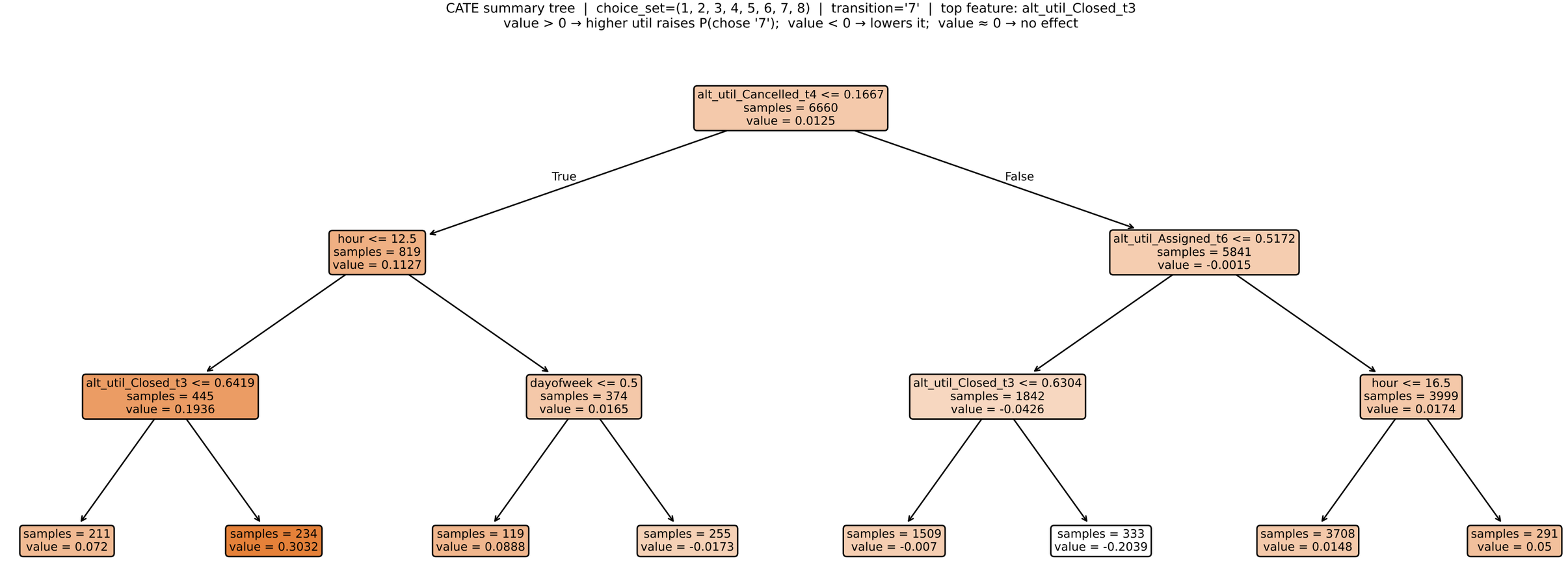



CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='5' | top feature: alt_util_Awaiting Assignment_t2
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t2 <= 0.5923
samples = 6660
value = 2.2171
True
False
alt_util_Wait_t1 <= 0.6256
samples = 3015
value = 7.6305
alt_util_Awaiting Assignment_t2 <= 0.6696
samples = 3645
value = -2.2608
hour <= 14.5
samples = 2060
value = 6.0516
dayofweek <= 0.5
samples = 955
value = 11.0365
hour <= 11.5
samples = 2459
value = 0.3543
alt_util_Unmatched_t7 <= 0.85
samples = 1186
value = -7.6827
samples = 1688
value = 7.1468
samples = 372
value = 1.0817
samples = 229
value = 14.9439
samples = 726
value = 9.804
samples = 1232
value = 2.726
samples = 1227
value = -2.0271
samples = 393
value = -2.8182
samples = 793
value = -10.0934


CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='6' | n=6660 | ATE=-1.3579 std=2.5073
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-1.3579
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='6' | top feature: alt_util_ln Progress_t5
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t5 <= 0.6349
samples = 6660
value = -1.3579
True
False
dayofweek <= 1.5
samples = 5123
value = -0.6703
alt_util_ln Progress_t5 <= 0.6756
samples = 1537
value = -3.6497
alt_util_Closed_t3 <= 0.6465
samples = 2024
value = 0.0357
alt_util_Awaiting Assignment_t2 <= 0.3185
samples = 3099
value = -1.1314
hour <= 14.5
samples = 941
value = -2.6441
alt_util_Awaiting Assignment_t2 <= 0.6518
samples = 596
value = -5.2375
samples = 1048
value = -0.8126
samples = 976
value = 0.9466
samples = 326
value = -2.6796
samples = 2773
value = -0.9494
samples = 782
value = -2.381
samples = 159
value = -3.938
samples = 284
value = -2.7759
samples = 312
value = -7.4782


CATE — choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='7' | n=6660 | ATE=0.0125 std=0.1138
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0125
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6, 7, 8) | transition='7' | top feature: alt_util_Closed_t3
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Cancelled_t4 <= 0.1667
samples = 6660
value = 0.0125
True
False
hour <= 12.5
samples = 819
value = 0.1127
alt_util_Assigned_t6 <= 0.5172
samples = 5841
value = -0.0015
alt_util_Closed_t3 <= 0.6419
samples = 445
value = 0.1936
dayofweek <= 0.5
samples = 374
value = 0.0165
alt_util_Closed_t3 <= 0.6304
samples = 1842
value = -0.0426
hour <= 16.5
samples = 3999
value = 0.0174
samples = 211
value = 0.072
samples = 234
value = 0.3032
samples = 119
value = 0.0888
samples = 255
value = -0.0173
samples = 1509
value = -0.007
samples = 333
value = -0.2039
samples = 3708
value = 0.0148
samples = 291
value = 0.05

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3, 4, 5, 6, 7, 8)	1	6660	-2.566352	2.798995	-11.388281	-4.46396	-2.386403	-0.567637	7.952292	18.1	81.9	alt_util_In Progress_t5
(1, 2, 3, 4, 5, 6, 7, 8)	2	6660	-2.229462	1.912742	-11.664036	-3.394926	-2.266943	-1.092141	6.36198	10.5	89.5	alt_util_Closed_t3
(1, 2, 3, 4, 5, 6, 7, 8)	3	6660	-1.625706	9.768329	-58.993276	-4.631718	-0.839406	2.433877	30.504693	44.1	55.9	alt_util_In Progress_t5
(1, 2, 3, 4, 5, 6, 7, 8)	4	6660	-0.016436	0.076828	-0.870808	-0.004759	-0.000137	0.000154	0.346378	32.6	67.4	alt_util_Unmatched_t7
(1, 2, 3, 4, 5, 6, 7, 8)	5	6660	2.217062	9.805686	-36.468528	-3.898848	3.048418	8.967881	32.650469	61.8	38.2	alt_util_Awaiting Assignment_t2
(1, 2, 3, 4, 5, 6, 7, 8)	6	6660	-1.357884	2.507341	-14.374836	-2.587352	-1.146645	0.114104	6.625985	26.7	73.3	alt_util_In Progress_t5
(1, 2, 3, 4, 5, 6, 7, 8)	7	6660	0.012513	0.113775	-0.804998	-0.00382	0.006146	0.025529	1.167855	67.6	32.4	alt_util_Closed_t3

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md